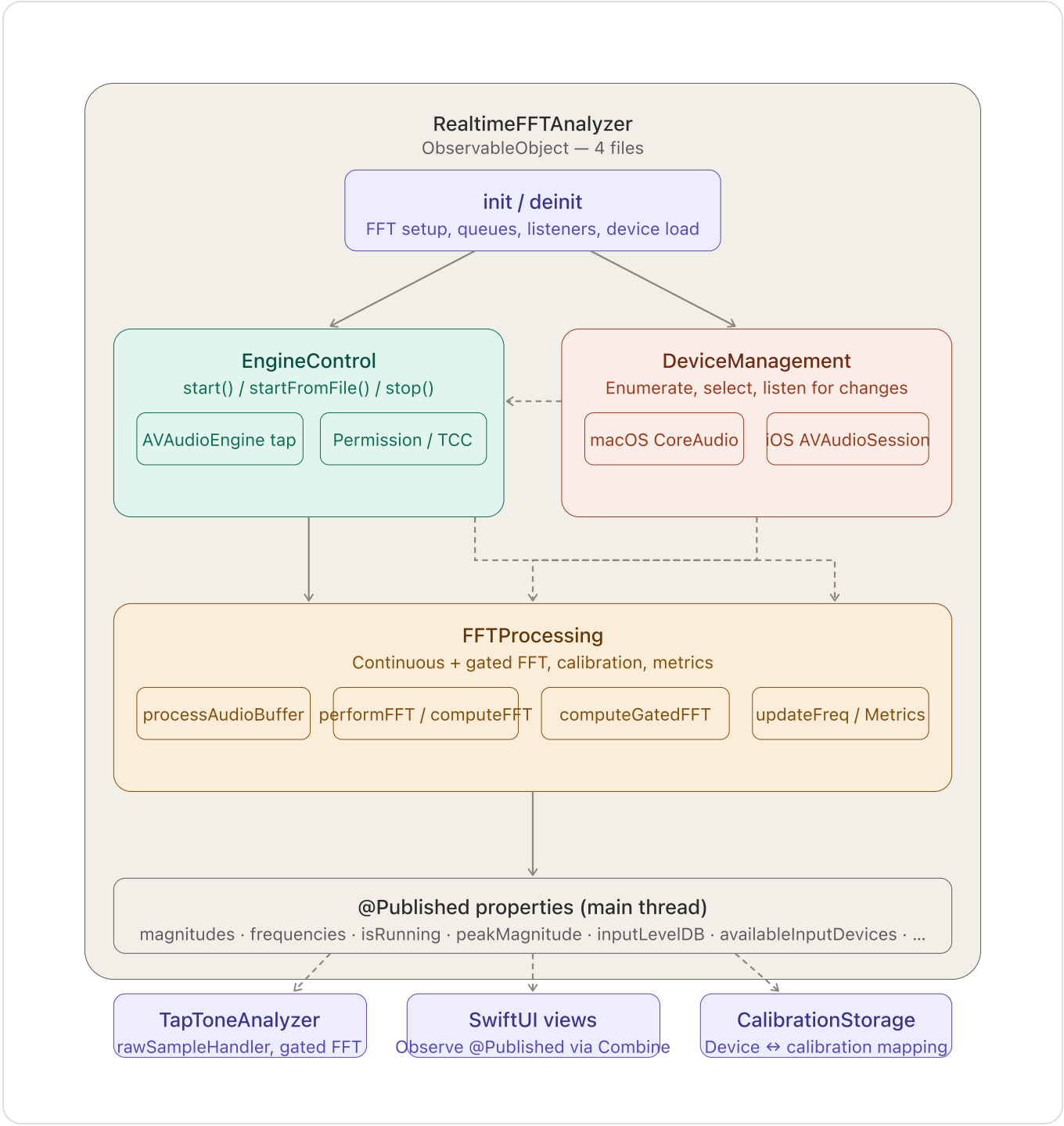
GuitarTap — RealtimeFFTAnalyzer Architecture

Generated from source analysis · 4 extension files

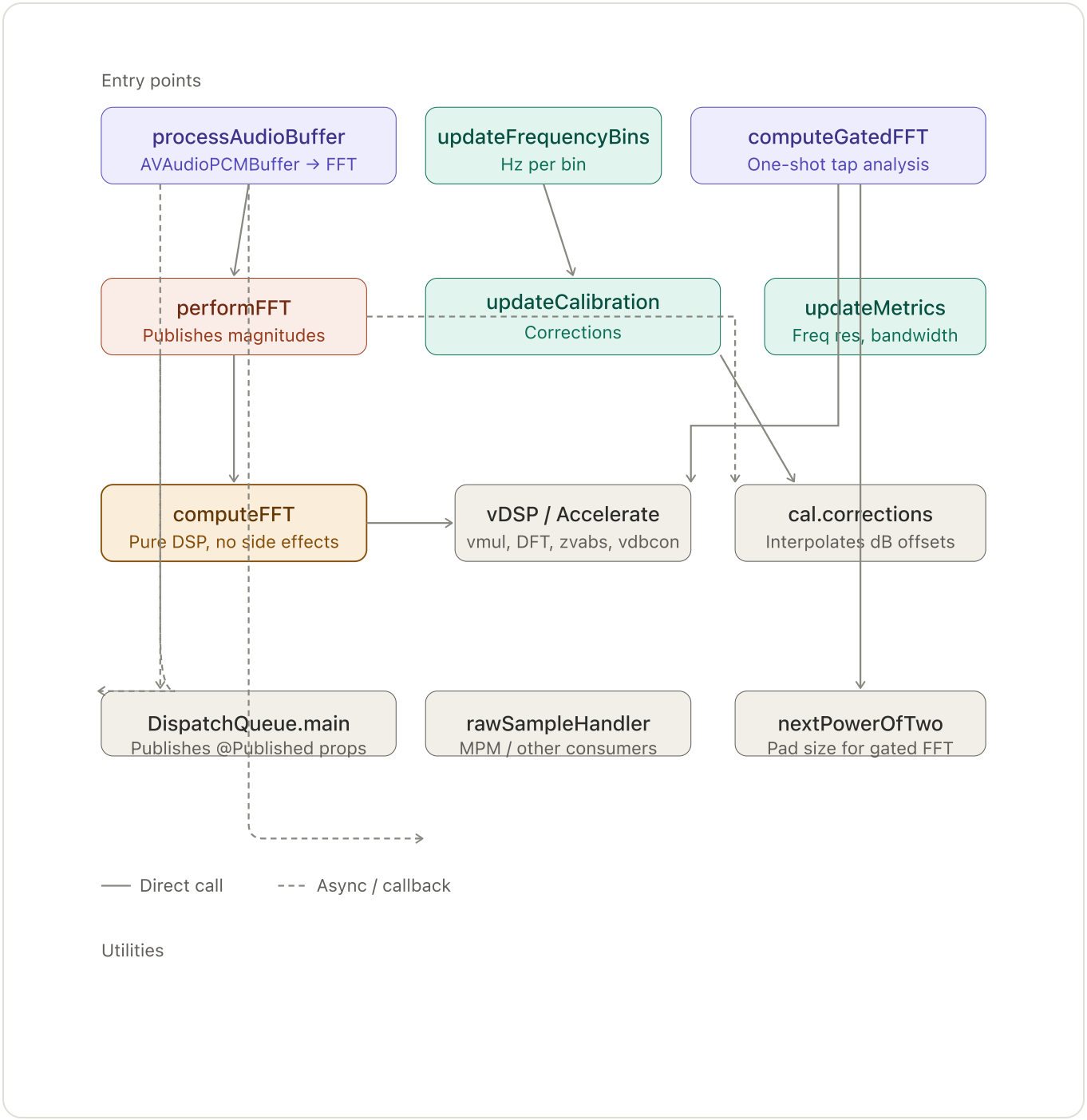
1. Full class architecture

Structural overview of all four files and how they relate. `init` bootstraps device enumeration and listeners. `EngineControl` and `DeviceManagement` both delegate into `FFTProcessing`, and all `@Published` output flows to SwiftUI and `TapToneAnalyzer` on the main thread.



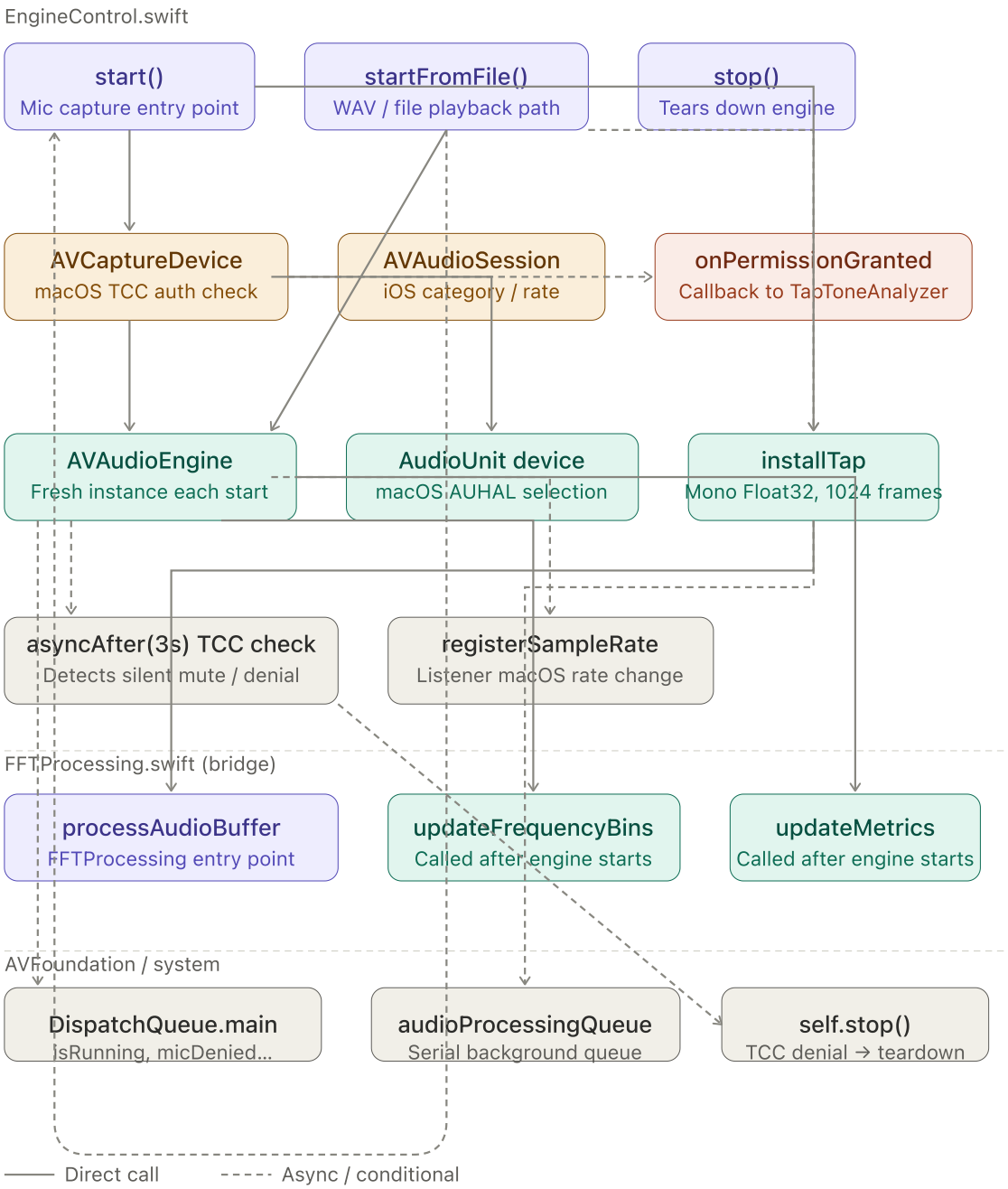
2. FFTProcessing call graph

`processAudioBuffer` is the streaming entry point — it accumulates samples, fires `rawSampleHandler` for `TapToneAnalyzer`, and dispatches chunks to `performFFT`. `performFFT` delegates pure math to `computeFFT` (no side effects), applies calibration, then publishes on the main thread. `computeGatedFFT` is a separate one-shot path for tap measurements using a Hann window.



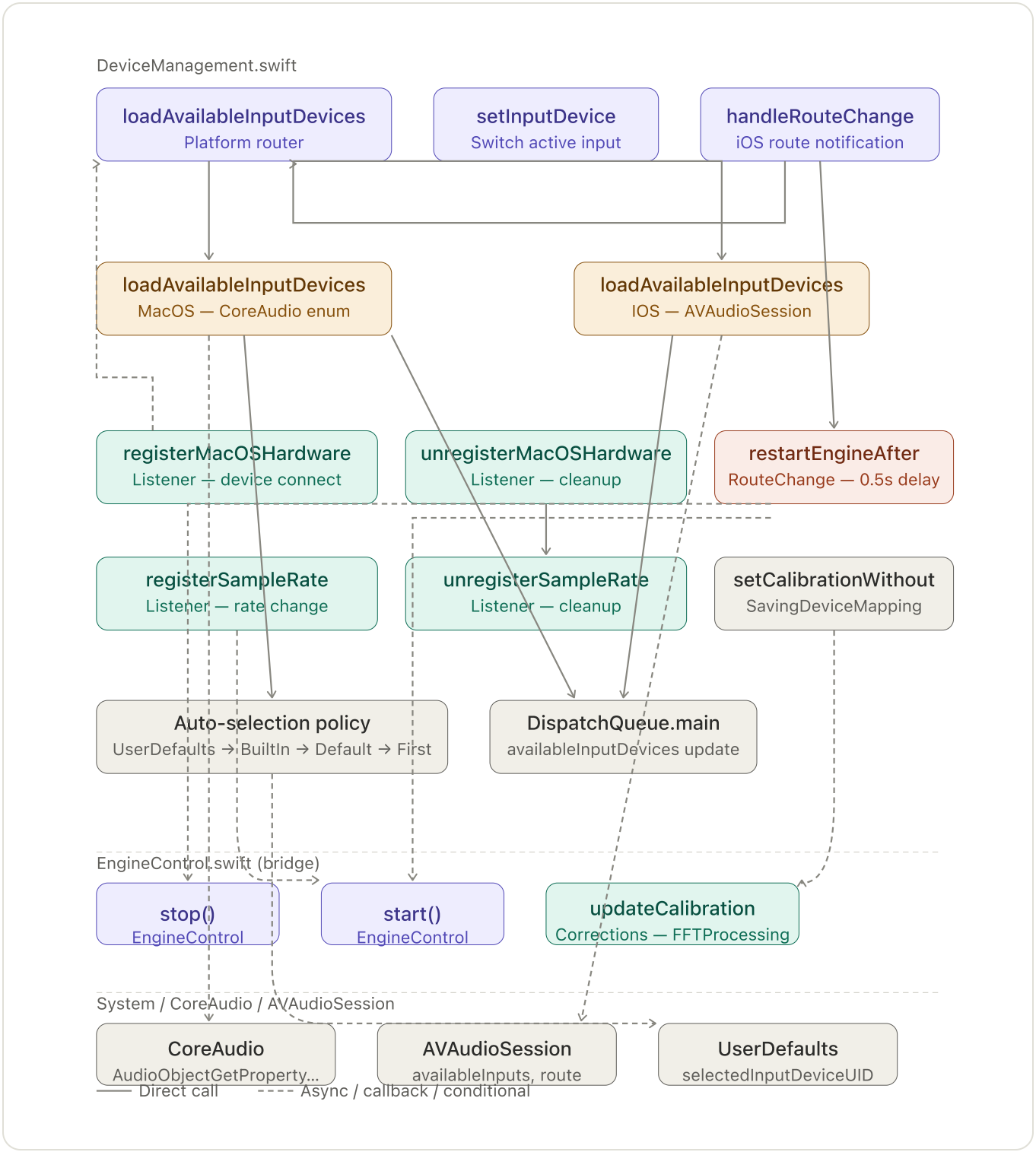
3. EngineControl call graph

`start()` and `startFromFile()` are the two public entry points. Both platform-split (macOS TCC / iOS AVAudioSession), set up `AVAudioEngine`, install a mono 1024-frame tap, then call `updateFrequencyBins` and `updateMetrics`. A 3-second async TCC recheck detects macOS silent-mute permission denial. `startFromFile()` loops back to `start()` when the file finishes playing.



4. DeviceManagement call graph

`loadAvailableInputDevices` is a platform router that forks into CoreAudio (macOS) or `AVAudioSession` (iOS). Two listener layers on macOS watch for device connect/disconnect and sample-rate changes. iOS uses `handleRouteChange + restartEngineAfterRouteChange`. All device switches ultimately call back into `EngineControl`'s `stop()` / `start()`.



Key architecture notes

- **Thread model:** All audio processing runs on `audioProcessingQueue` (serial, userInteractive). Buffer access is guarded by `bufferAccessQueue`. All `@Published` updates dispatch to the main thread.
- **Two FFT paths:** Continuous path uses a rectangular window at ~2 Hz for live display. Gated path (`computeGatedFFT`) uses a Hann window for tap-tone plate/brace measurements.

- **macOS TCC workaround:** A 3-second delayed check detects the "silent all-zero stream" condition that occurs when macOS caches `.authorized` after a permission revocation.
- **Platform split:** Device enumeration, session setup, and route-change handling differ entirely between macOS (CoreAudio / AUHAL) and iOS (AVAudioSession). Both paths converge at `installTap` and `processAudioBuffer`.
- **Calibration loop prevention:** `isLoadingDeviceCalibration` and `isApplyingDeviceSwitch` guard flags break `didSet` feedback loops in `selectedInputDevice` and `activeCalibration`.
- **File playback:** `startFromFile()` routes the same pipeline as the mic tap — same `processAudioBuffer → performFFT` path — and calls `start()` when the file finishes to switch back to live mic.

Generated by Claude from GuitarTap source analysis · RealtimeFFTAnalyzer.swift + 3 extensions